Anastasiia Tkachenko

Kahlert School of Computing, University of Utah

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RESEARCH INTERESTS

Theoretical Computer Science, Algorithms and Data Structures, Computational Geometry, Algorithm Design in Interdisciplinary Research

EDUCATION

Ph.D. in Computer Science (expected May 2027), University of Utah, Salt Lake City, Utah **2023-Present**
Advisor: Professor Haitao Wang
GPA: 4.0

Doctoral Coursework in Computer Science, Utah State University, Logan, Utah **2021-2023**
GPA: 4.0

M.S. in Applied Mathematics and Informatics, Novosibirsk State University, Novosibirsk, Russia **2016-2018**
Diploma with highest honors (summa cum laude)
Thesis: Maximization of labor productivity in a two-level optimal control problem: perturbed production function

B.S. in Applied Mathematics and Informatics, Novosibirsk State University, Novosibirsk, Russia **2012-2016**
Diploma with highest honors (summa cum laude)

HONORS, AWARDS, AND GRANTS

Graduate Student Professional Development Supplement Award, The University of Utah ($500) **2026**

CG Young Researchers Forum Travel Grant, Rutgers University ($500) **2026**

Fall Workshop on Computational Geometry Travel Grant, Tufts University ($1,000) **2024**

Outstanding Computer Science Graduate Teaching Assistant, Utah State University ($100) **2023**

Graduate Student Enhancement Scholarship, Utah State University ($4000) **2022-2023**

Personal Fellowship of Sberbank - 12UP, National competition ($3000) **2017-2018**

Second-Degree Diploma, International Scientific Student Conference **2017**

Scholarship for Scientific Achievements, Novosibirsk State University ($1500) **2017**

Participant, The Russian Foundation for Basic Research Grant: Mathematical analysis of dynamic models of non-classical markets ($2500) **2016-2018**

Participant, The Russian Foundation for Basic Research Grant: Analysis of complex socio-economic systems: new approaches and methods ($1000) **2016-2018**

PUBLICATIONS

Journal Articles

1. **Anastasiia Tkachenko** and Haitao Wang. "Computing Maximum Cliques in Unit Disk Graphs." *Computational Geometry: Theory and Applications*, 135, Article 102269, pp. 1-12, 2026. DOI.
2. Hui Wang, Jesse B. Brown, Caulder Council, **Anastasiia Tkachenko**, Yuqin Qian, Lizheng Xue, Zhiqiang Li, Xiaojun Qi, and Yi Rao. "Theory and Experiments of Angle-Dependent Second-Order Nonlinear Optical Scattering." *Physical Review B*, 2026. DOI.
3. Nikolay Tkachenko, **Anastasiia Tkachenko**, Benjamin Nebgen, Sergei Tretiak, and Alexander I. Boldyrev. "Neural Network Atomistic Potentials for Global Energy Minima Search in Carbon Clusters." *Physical Chemistry Chemical Physics*, 25, 21173-21182, 2023. DOI.
4. Vladimir Kulyukin, Daniel Coster, **Anastasiia Tkachenko**, Daniel Hornberger, and Aleksey Kulyukin. "Ambient Electromagnetic Radiation as a Predictor of Honey Bee (Apis mellifera) Traffic in Linear and Non-Linear Regression: Numerical Stability, Physical Time and Energy Efficiency." *Sensors*, 23, 2584, 2023. DOI.
5. Vladimir Kulyukin, **Anastasiia Tkachenko**, Kristoffer Price, William Meikle, and Milagra Weiss. "Integration of Scales and Cameras in Nondisruptive Electronic Beehive Monitoring: On the Within-Day Relationship of Hive Weight and Traffic in Honeybee (Apis mellifera) Colonies in Langstroth Hives in Tucson, Arizona, USA." *Sensors*, 22, 4824, 2022. DOI.
6. Nikolay Tkachenko, **Anastasiia Tkachenko**, Vladimir Kulyukin, and Alexander I. Boldyrev. "DFT Study of Microsolvated $[NO_3 \cdot (H_2O)_n]^-$ (n = 1-12) Clusters and Molecular Dynamics Simulation of Nitrate Solution." *The Journal of Physical Chemistry A*, 125, 8899, 2021. DOI.

7. Vladimir Kulyukin, Nikhil Ganta, and **Anastasiia Tkachenko**. "On Image Classification in Video Analysis of Omnidirectional Apis Mellifera Traffic: Random Reinforced Forests vs. Shallow Convolutional Networks." *Applied Sciences*, 11(17), 8141, 2021. DOI.

Peer-reviewed Conference Papers

1. **Anastasiia Tkachenko** and Haitao Wang. "Maximum Independent Sets in Disk Graphs with Disks in Convex Position." *Proceedings of the 20th Scandinavian Symposium on Algorithm Theory (SWAT)*, pp. 40:1-40:18, Copenhagen, Denmark, June 2026. DOI.
2. **Anastasiia Tkachenko** and Haitao Wang. "Computing Dominating Sets in Disk Graphs with Centers in Convex Position." *Proceedings of the 17th Latin American Theoretical Informatics Symposium (LATIN)*, Florianópolis, Brazil, April 2026. DOI.
3. **Anastasiia Tkachenko** and Haitao Wang. "Computing Maximum Cliques in Unit Disk Graphs." *Proceedings of the 37th Canadian Conference on Computational Geometry (CCCG)*, pp. 283-291, Toronto, Canada, August 2025. DOI.
4. **Anastasiia Tkachenko** and Haitao Wang. "Dominating Set, Independent Set, Discrete k-Center, Dispersion, and Related Problems for Planar Points in Convex Position." *Proceedings of the 42nd International Symposium on Theoretical Aspects of Computer Science (STACS)*, pp. 73:1-73:20, Jena, Germany, March 2025. DOI.

- **ADDITIONAL CONFERENCE ABSTRACTS (not formal publications)**

Earlier publications appeared under the name Anastasiia Kotova.

1. **Anastasiia Tkachenko** and Haitao Wang. "Maximum Independent Sets in Disk Graphs with Disks in Convex Position." *Proceedings of the Computational Geometry: Young Researchers Forum (CG:YRF),* CG Week 2026, Rutgers University, New Brunswick, New Jersey, June 2-5, 2026.
2. **Anastasiia Tkachenko** and Haitao Wang. "Computing Dominating Sets in Disk Graphs with Centers in Convex Position." *Proceedings of the 32nd Annual Fall Workshop on Computational Geometry (FWCG),* Queens, New York, November 7-8, 2025.
3. **Anastasiia Tkachenko** and Haitao Wang. "Independent Set and Dispersion Problems for Planar Points in Convex Position." *Proceedings of the 31st Annual Fall Workshop on Computational Geometry (FWCG)*, Tufts University, Medford, Massachusetts, November 15-16, 2024.
4. **Anastasiia Tkachenko**. "On a Special Rational Control Mode with Perturbation of the Production Function." *Proceedings of the 56th International Scientific Student Conference*, p. 105, Novosibirsk State University, Novosibirsk, Russia, March, 2018.
5. **Anastasiia Kotova**. "On the Influence of Investors' Propensity to Consume on the Rate of Personal Income Tax." *Proceedings of the International Conference Dedicated to the 60th Anniversary of the Sobolev Institute, Mathematics in the Modern World*, p. 544, Novosibirsk State University, Novosibirsk, Russia, August 2017.
6. **Anastasiia Kotova**. "On Maximizing Labor Productivity in a Two-Level Optimal Control Problem." *Proceedings of the 55th International Scientific Student Conference*, p. 197, Novosibirsk State University, Novosibirsk, Russia, April, 2017.
7. **Anastasiia Kotova**. "On Maximizing Labor Productivity in the Modified Ramsey-Solow Model." Proceedings of the 54th International Scientific Student Conference, p. 215, Novosibirsk State University, Novosibirsk, Russia, March, 2016.

CONFERENCE AND SEMINAR PRESENTATIONS

1. "Maximum Independent Sets in Disk Graphs with Disks in Convex Position," *the 20th Scandinavian Symposium on Algorithm Theory (SWAT)*, the Technical University of Denmark, Copenhagen, Denmark, June 17-19, 2026.
2. "Computing Dominating Sets in Disk Graphs with Centers in Convex Position*," the 17th Latin American Theoretical Informatics Symposium (LATIN)*, Universidade Federal de Santa Catarina, Florianópolis, Brazil, April 12-17, 2026.
3. "Maximum Independent Sets in Disk Graphs with Disks in Convex Position*," Computational Geometry: Young Researchers Forum (CG:YRF),* CG Week 2026, Rutgers University, New Brunswick, New Jersey, June 2-5, 2026.
4. "Computing Maximum Cliques in Unit Disk Graphs," *the 37th Canadian Conference on Computational Geometry (CCCG),* York University, Toronto, Canada, August 13-15,2025.
5. "Dominating Set, Independent Set, Discrete k-Center, Dispersion, and Related Problems for Planar Points in Convex Position," *the 42nd International Symposium on Theoretical Aspects of Computer Science (STACS)*, University of Jena, Jena, Germany, March 4-7, 2025.
6. "Maximum Dominating Sets in Disk and Unit-Disk Graphs with Disk Centers in Convex Position." *Theory Group Seminars*, University of Utah, Salt Lake City, Utah, November 26, 2025.
7. "Computing Dominating Sets in Disk Graphs with Centers in Convex Position*." 32nd Annual Fall Workshop on Computational Geometry (FWCG)*, Queens College, City University of New York, New York, November 7-8, 2025.

8. "Independent Set and Dispersion Problems for Planar Points in Convex Position." *31st Annual Fall Workshop on Computational Geometry (FWCG)*, Tufts University, Medford, Massachusetts, November 15-16, 2024.
9. "Independent Set Problem for Planar Points in Convex Position." *Theory Group Seminars*, University of Utah, Salt Lake City, Utah, October 25, 2024.

RESEARCH EXPERIENCE

Research Assistant, Kahlert School of Computing, The University of Utah **2023-2026**
- Designed and analyzed algorithms for domination, independent set, clique, dispersion, and k- center problems in geometric intersection graphs.
- Developed structural results for points and disks in convex position that enable efficient algorithms.

Research Assistant, Utah State University **May-Aug 2023**

Departments: Computer Science; Chemistry and Biochemistry
- Developed the theoretical framework for sum-frequency-generation scattering spectroscopy and prepared manuscript materials.

Research Assistant, Novosibirsk State University **2016-2018**
- Developed mathematical models and established their validity and applicability, conducted literature reviews and prepared materials for grant submissions on mathematical models of government-investor economic processes.
- Research resulted in four conference papers and abstracts.

TEACHING EXPERIENCE

Teaching Assistant, The University of Utah **2024-Present**
- Computational Geometry (40 students), Teaching Assistant, Fall 2026
- Algorithms (CS 4150) (200 students), Lead Teaching Assistant, Spring 2026; Teaching Assistant, Spring 2024
- Models of Computation (CS 3100) (200 students), Lead Teaching Assistant, Spring 2025

Led discussion sections; graded homework, programming assignments, exams, and projects; supervised and coordinated the teaching-assistant team as lead TA.

Teaching Assistant, Utah State University **2021-2023**
- Programming Languages (CS 4700) (50 students), delivered several course lectures, Spring 2023
- Advanced Algorithms (CS 5050) (50 students), delivered several course lectures, Fall 2022
- Scientific Computing (CS 3430) (50 students), Spring 2022
- Theory of Computability (CS 5000) (50 students), Fall 2021
- Introduction to Computer Science (CS 1400) (200 students), Spring 2021

Prepared lessons and instructional materials, facilitated discussion sections, and graded coursework and exams

Private Tutor, Mathematics, Novosibirsk State University **2015-2019**
- Calculus
- Mathematical Statistics
- Probability Theory

Provided individualized instruction, developed tailored lesson plans and practice materials to support students with varying levels of preparation, developed tailored lesson plans and practice materials to support students with varying levels of preparation, helped students prepare for examinations and strengthen their problem-solving skills.

SERVICE AND LEADERSHIP

Peer Reviewer, International Symposium on Computational Geometry (SoCG); European Symposium on Algorithms (ESA); Engineering Optimization **2025-2026**

Conference Volunteer, ACM STOC 2026, Salt Lake City, Utah **June 2026**

President, Computer Science Graduate Student Council, Utah State University **2022-2023**

Vice President, Association for Computing Machinery for Women, Utah State University **2021-2023**